

Ideally, you should opt for laptops with memory modules that have just one Rank. This is with regards to DDR4 memory modules which form the vast majority of configurations. As for the capacity of memory needed, most games require about 16 GB of memory. Whereas day-to-day productivity applications don't need more than 8 GB of memory. Certain use cases such as content creation applications would require much more memory, but there are usually specially configured creator laptops for those use-cases. If you're torn between two laptops with just the memory capacities being different then you're usually better off getting the cheaper unit and adding memory rather than getting the higher-priced model. This remains true as long as the laptop doesn't have soldered RAM such as in Apple Macbooks. There is no upgradability in such cases.

3. GRAPHICS CARD

The graphics card isn't just meant for gaming. Laptops might have integrated graphics or discrete graphics or a mix of both. Certain applications benefit greatly from a graphics card that supports OpenGL / OpenCL. Content creator applications and



Cooling is essential for gaming laptops

software use for CAD/CAM applications also leverage the graphics card. However, you should check the specifications page of that software to understand whether the graphics card present in the laptop can actually be used. Usually, there are separate graphics cards from AMD and NVIDIA such as the Quadro and Radeon Pro lineups which calculate figures at a higher accuracy. Most

folks don't need the unique features that such cards provide and can easily make do with consumer-grade graphics cards such as the GeForce and Radeon lineups.

As for gaming, you will get better performance with better GPUs. However, not all GPUs are the same. Laptops manufacturers have the freedom to vary the TDP (Thermal Design Power) as per the cooling they've included in the laptop. As a result, non-gaming lineups usually have the GPUs set to a lower TDP since the cooling isn't all that great. So even if you have a mainstream laptop with an RTX 3070, unless the TDP is set to the maximum level, you will have reduced performance. You can check the TDP of the mobile GPUs in the driver control panel and that can be compared against the official numbers from NVIDIA or AMD.

4. STORAGE

Most laptops these days ship with an SSD and the higher-end laptops usually have NVMe SSDs which are much faster. You don't necessarily have to get the fastest NVMe SSD. Anything that provides 2000 MBps or higher should be more than sufficient. Most folks can't even tell the difference when they're using an NVMe SSD with higher speeds. That's because the higher spec SSDs are beneficial only when you're transferring very large files. Whereas when an operating system is running off an SSD, the most frequently used or accessed files are much smaller. However, always opt for a laptop that has an SSD. The difference between a laptop that has the OS on a hard drive and one with the OS on an SSD is night and day!

Then there's the expandability factor. If you've got an additional M.2 slot for NVMe storage then you can add additional storage space later on. However, this is not clearly mentioned on most laptop's specification pages. And there's no easy way to tell if your laptop has one without

opening it up. The best you can do is check YouTube for disassembly videos of the same laptop model that is selling in India. Why India? Because the chassis used for the laptops varies from country to country and in some rare cases, it will even vary between the marketplaces! If the exact model number is the same, then you'd need not worry otherwise, there's a slight chance that you might not get the same configuration.



An additional NVMe slot will help

5. WI-FI 6/E

It goes without saying that wireless connectivity is extremely important on portable devices. Wi-Fi 6E is the latest wireless standard, however, we don't see any Wi-Fi 6E routers in India. Wi-Fi 6E paves way for a new 6 GHz band to be used for wireless data transmission allowing for much higher throughput, but the 6 GHz frequency band is not open to the public in India. The government has to open up the band for the consumers, only then will we see router manufacturers bring their latest equipment to the country. For now, you can make do with Wi-Fi 6 routers which are pretty great on their own.

There aren't any laptops with Wi-Fi 6E even though the latest processors from Intel and AMD have added support for the same. You might see laptops with Wi-Fi 6E hit the market very soon, but that shouldn't be a factor in your purchase decision.

6. DISPLAY PANEL

Display panels have a huge impact on the way you experience content on the laptop. A lot of the high-tier gaming laptops now ship with high refresh-rate IPS-like panels. IPS panels are a bit expensive so manu-